

EXPERIMENT (14)

Ex. No.: 14

Date:

AIM

To learn how to use TCL commands COMMIT, SAVEPOINT and ROLLBACK in the Hostel Management System database.

DESCRIPTION

Transaction Control Language (TCL) commands manage transactions in the database. COMMIT permanently saves changes, SAVEPOINT creates a point to roll back to, and ROLLBACK restores data to a previous state.

SYNTAX

COMMIT;

ROLLBACK;

ROLLBACK TO savepoint_name;

SAVEPOINT savepoint_name;

Problem

Create transactions on the Hostel_Student table, create savepoints between INSERT operations, roll back to a savepoint, and verify the output using SELECT statements.

Program

USE HostelDB;

START TRANSACTION;

```
INSERT INTO Hostel_Student
(StudentID, StudentName, Gender, DOB, MobileNo, RoomID, Email)
VALUES
(301,'Rakesh','M','2005-01-15',9876543001,101,'rakesh@gmail.com');
```

SAVEPOINT A;

```
INSERT INTO Hostel_Student
(StudentID, StudentName, Gender, DOB, MobileNo, RoomID, Email)
VALUES
(302,'Sneha','F','2005-02-20',9876543002,102,'sneha@gmail.com');
```

SAVEPOINT B;

INSERT INTO Hostel_Student

(StudentID, StudentName, Gender, DOB, MobileNo, RoomID, Email)

VALUES

(303,'Vignesh','M','2005-03-12',9876543003,103,'vignesh@gmail.com');

SAVEPOINT C;

SELECT * FROM Hostel_Student;

ROLLBACK TO B;

SELECT * FROM Hostel_Student;

COMMIT;

SELECT * FROM Hostel_Student;

The screenshot shows a SQL IDE window with a script editor and a result grid. The script editor contains the following SQL commands:

```
113 VALUES
114 (402,'Sneha','F','2005-02-20',9876543002,102,'sneha@gmail.com');
115 SAVEPOINT B;
116 INSERT INTO Hostel_Student
117 (StudentID, StudentName, Gender, DOB, MobileNo, RoomID, Email)
118 VALUES
119 (403,'Vignesh','M','2005-03-12',9876543003,103,'vignesh@gmail.com');
120 SAVEPOINT C;
121 SELECT * FROM Hostel_Student;
```

The result grid displays the following data:

StudentID	StudentName	Gender	DOB	MobileNo	RoomID	Email
201	Lokesh	M	2005-03-15	9876543210	101	lokesh@gmail.com
202	Rahul	M	2004-07-10	9876543211	102	rahul@gmail.com
401	Rakesh	M	2005-01-15	9876543001	101	rakesh@gmail.com
402	Sneha	F	2005-02-20	9876543002	102	sneha@gmail.com
403	Vignesh	M	2005-03-12	9876543003	103	vignesh@gmail.com
NOTE	NOTE	NOTE	NOTE	NOTE	NOTE	NOTE

RESULT

Thus, the TCL commands COMMIT, SAVEPOINT and ROLLBACK were successfully implemented on the Hostel Management System database.